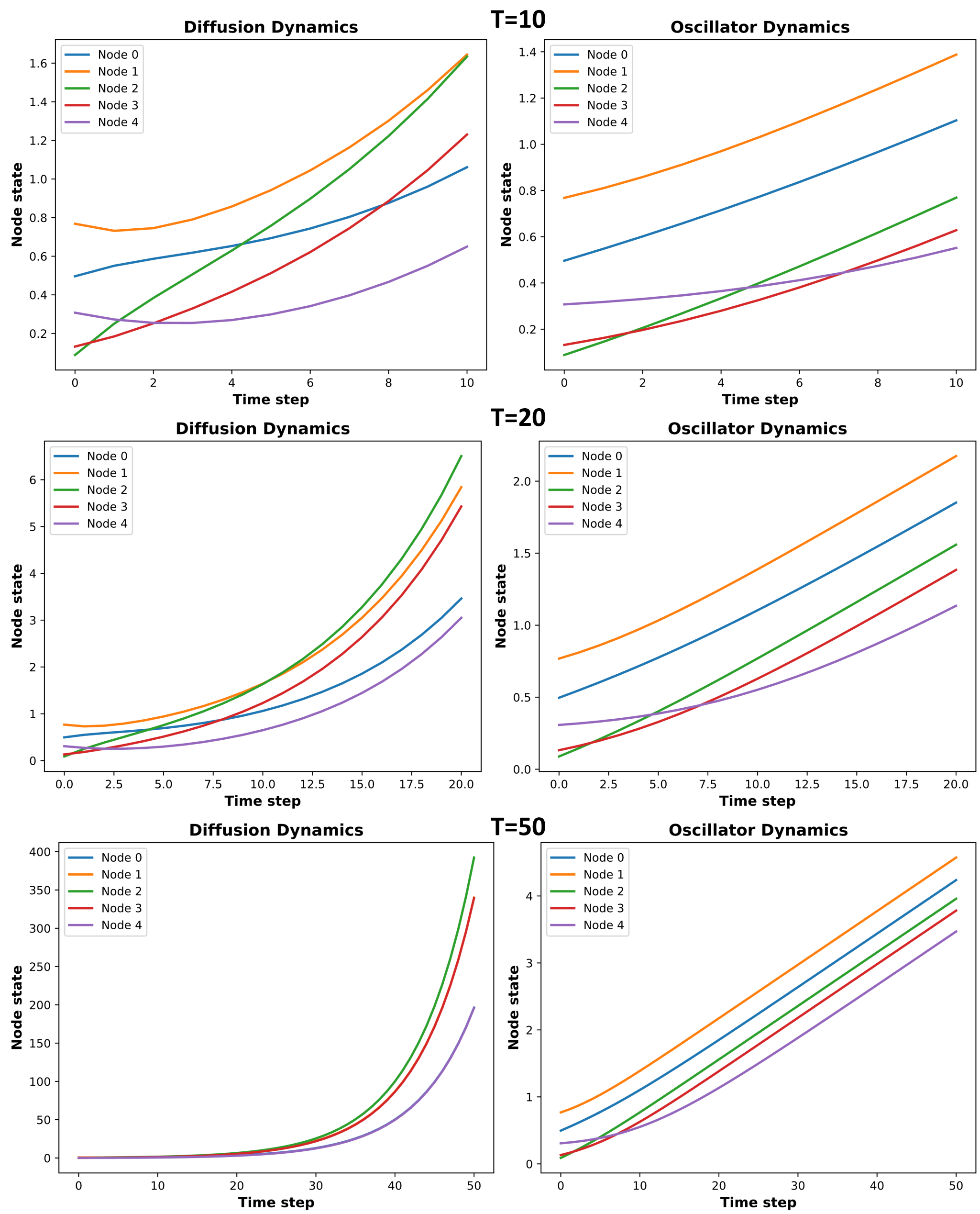

Figure 7. Comparison of node trajectories under prolonged simulation (T=10, T=20, T=50). In diffusion-based dynamics (left), all node representations converge to a single trajectory, reflecting the oversmoothing effect common in deep GNNs. In contrast, our oscillator-based dynamics (right) maintain distinct trajectories across nodes, demonstrating preserved discriminability due to nonlinear feedback modulation.

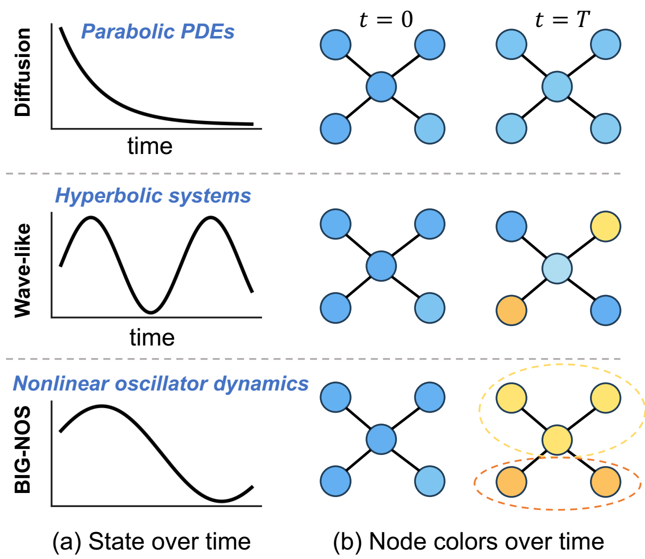

Figure 8. Illustration comparing the dynamics of diffusion-based GNNs, wave-like systems, and our proposed *BIG-NOS* model. (a) Temporal evolution of a node's state under each model. Diffusion leads to rapid smoothing, erasing local distinctions. Wave-like systems propagate structured oscillations, preserving local variations over time. *BIG-NOS* exhibits nonlinear synchronization dynamics that retain both temporal structure and topological distinctions. (b) Toy graph visualization of node states at $t = 0$ and $t = T$. Diffusion homogenizes node states, blurring community boundaries. Wave-like dynamics retain rougher spatial patterns. In contrast, *BIG-NOS* maintains subnetwork-level distinctions through phase-aligned but non-averaged interactions, preserving community-level structure. Although not explicitly hyperbolic, *BIG-NOS* shares essential properties with hyperbolic (wave-like) systems—preserving local variations and structured node information.

Table 9. [Computational overhead] Inference time and accuracy of our *BIG-NOS* under different configurations of L and Q (Cora dataset).

L	1 (params=41.2M)						2 (params=50.7M)					4 (params=69.5M)				8 (params=107.3M)	
Q	4	8	16	32	64	128	4	8	16	32	64	4	8	32	64	8	16
layers	4	8	16	32	64	128	8	16	32	64	128	16	32	64	128	64	128
Inference Time (ms)	39.7	44.4	46.9	48.1	53.9	76.3	41.5	49.7	51.8	63.8	85.1	46.4	53.5	74.9	94.7	75.5	99.3
Acc (%)	81.0	80.4	81.4	81.9	91.9	81.9	-	-	-	-	-	-	-	-	-	-	-

Table 10. Inference time and accuracy of GCN under different layer depths.

GCN	**layers**	4	8	16	32	64	128
	Inference Time (ms)	20.7	42.5	49.7	64.9	85.8	125.8
	Params (M)	1.8	2.8	4.9	9.2	17.6	34.4
Acc (%)		80.86	75.70	30.54	29.94	31.48	30.4